

From Kirkwood Rough

PRINCIPLES OF RADIO COMMUNICATION

BY

JOHN H. MORECROFT

*Professor of Electrical Engineering, Columbia University
Past President Institute of Radio Engineers*

ASSISTED BY

A. PINTO

Electrical Engineer, Otis Elevator Co.

AND

W. A. CURRY

Assistant Professor of Electrical Engineering, Columbia University

SECOND EDITION—THOROUGHLY REVISED

TOTAL ISSUE TWELVE THOUSAND

To convert Inductance expressed in cm to henrys
multiply by 10^{-9} or uh $\text{cm} \cdot 10^{-3}$

NEW YORK

JOHN WILEY & SONS, INC.

LONDON: CHAPMAN & HALL, LIMITED

1927

To convert Capacitance expressed in cm to pf
divide by .9

A radio topographic map of New York City. The figures give the "millivolts per meter" of the carrier wave when not modulated. The station was located among the tall steel buildings in lower Manhattan. (For other radio topographic maps see Journal A.I.E.E., Jan., 1927.)

Frontispiece.



TC
copy

Kirkwood Rough 12-13-63

$$\frac{\text{cm}}{.9} = \text{picofarads (capacitance)}$$

$$\text{cm} \times 10^{-3} = \text{microhenries (inductance)}$$

$$\text{cm} \times 10^{-9} = \text{henries}$$

Did he send capacitor section?

COEFFICIENT OF SELF-INDUCTION

193

They are all based on a uniform distribution of current in the cross-section of the conductor; in case the current is unevenly distributed, due to skin effect, etc., hyperbolic functions are generally required for an accurate formula.

For exact formulae the student should consult the various publications on the subject, notably those of the Bureau of Standards.

Self-induction of a Single Straight Vertical Wire Distant from All Other Conductors.—

$$L = 2l \left(\log \frac{2l}{r} - \frac{3}{4} \right) \text{ cm., (9)}$$

where l = length of wire in cm.;

r = radius of wire in cm.;

L = coefficient of self-induction in cm.;

$\log$ = logarithm to the base e , as it is for all the succeeding formulae.

Equation (9) assumes the material of the wire to have a permeability of unity. For uniform current distribution, and permeability differing from unity,

$$L = 2l \left(\log \frac{2l}{r} - 1 + \frac{\mu}{4} \right) \text{ cm., (10)}$$

where μ = the value of the permeability.

For a Single Horizontal Wire.—

$$L = 2l \left(\log \frac{2h}{r} + \frac{1}{4} \right) \text{ cm., (11)}$$

where l = length in cm.;

r = radius of wire in cm.;

h = height of wire, above earth, in cm.

For a Single Circular Turn of Round Wire.—

$$L = 4\pi R \left[\left(1 + \frac{r^2}{8R^2} \right) \log \frac{8R}{r} + \frac{r^2}{24R^2} - 1.75 \right] \text{ cm., . . . (12)}$$

where R = radius of turn, to center of conductor, in cm.;

r = radius of cross-section of conductor.

For a Single Layer Solenoid, Closely Wound.—

$$L = 4\pi^2 R^2 n_1^2 l K \text{ cm., (13)}$$

where R = radius of coil, to center of wire, in cm.;
 n_1 = number of turns of wire per cm. length;
 l = length of winding in cm.

K = summation of a certain series, which series depends upon the form of the coil. These series have been summed by H. Nagaoka and are given in Table V. The value of K is given in terms of $\frac{2R}{l}$, i.e., the ratio of the coil diameter to the coil length.

TABLE V

$\frac{2R}{l}$	Diameter Length	K	Diameter Length	K
.00		1.000	.95	.700
.05		.979	1.00	.688
.10		.959	1.10	.667
.15		.939	1.20	.648
.20		.920	1.40	.611
.25		.902	1.60	.580
.30		.884	1.80	.551
.35		.867	2.00	.526
.40		.850	2.50	.472
.45		.834	3.00	.429
.50		.818	3.50	.394
.55		.803	4.00	.365
.60		.789	4.50	.341
.65		.775	5.00	.320
.70		.761	6.00	.285
.75		.748	7.00	.258
.80		.735	8.00	.237
.85		.723	9.00	.219
.90		.711	10.00	.203

The values of K given in the table assume a current distribution uniform throughout the conductor, and so give too large a value of L , if, due to skin effect, the current concentrates in the inner side of the winding. The decrease in inductance due to this effect is shown in Table VI, in which are tabulated the experimentally determined inductances and resistances of the two edgewise-wound ribbon coils referred to on p. 163, and pictured in Fig. 53. It may be found by calculation from the figures given that at high frequencies the current is practically concentrated in the inner side of the coil.

Best Form of Solenoid.—It may be seen from a few calculations, using Eq. (13), that a given amount of wire, to be wound into a single-

layer solenoid, should have to be obtained. This occurs at a certain length. The variation of L and R with length, however; thus if the ratio of L to R is fixed (for fixed length of wire),

RESISTANCE AND INDUCTANCE

Frequency in 10^3 cycles.....	.043	
L in 10^{-6} henry....	489	
R in 10^{-3} ohm.....	13	
Frequency in 10^3 cycles.....	7.00	16
L in 10^{-6} henry....	458	
R in 10^{-3} ohm.....	295	

Frequency in 10^3 cycles.....	.043	.1
L in 10^{-6} henry....	613	6
R in 10^{-3} ohm.....	23	2
Frequency in 10^3 cycles.....	3.50	6
L in 10^{-6} henry....	581	5
R in 10^{-3} ohm.....	245	4

Two-layer solenoids, on the other hand, are feasible for radio work, as coils are sometimes used, having "banked" winding. Multi-layer solenoids must be built in such a way as to be described on p. 164.

layer solenoid, should have a certain form if the maximum inductance is to be obtained. This occurs when the diameter is 2.45 times the coil length. The variation of L with departure from this form is not great, however; thus if the ratio is made as low as 1.5 or as high as 4.5 the decrease in L (for fixed length of wire), is only 3 per cent.

TABLE VI
RESISTANCE AND INDUCTANCE OF EDGEWISE-WOUND RIBBON COILS
Coil No. 1

Frequency in 10^3 cycles.....	.043	.088	.128	.248	.338	.450	.730	1.250	3.50
L in 10^{-6} henry....	489	485	482	476	472	470	466	464	460
R in 10^{-2} ohm.....	13	15	19	26	31	36	46	72	176
Frequency in 10^3 cycles.....	7.00	16.4	25.2	50.0	75.0	100	125	150	
L in 10^{-6} henry....	458	455	452	451	454	457	456	460	
R in 10^{-2} ohm.....	295	725	945	1345	1775	2205	2745	3440	

Coil No. 2

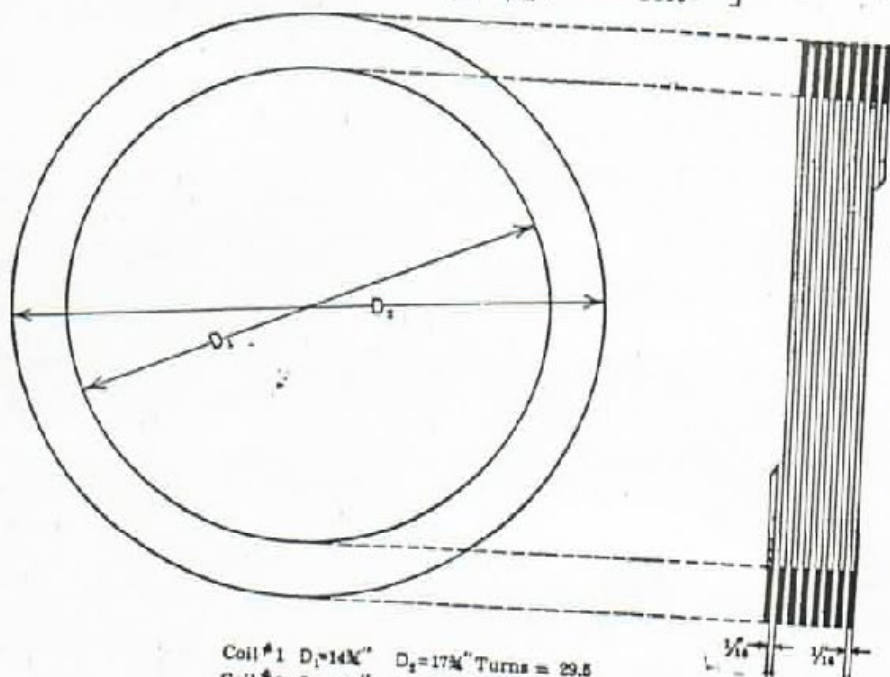
Frequency in 10^3 cycles.....	.043	.100	.150	.200	.300	.400	.600	1.000	1.60	2.44
L in 10^{-6} henry....	613	608	604	602	598	595	592	585	585	583
R in 10^{-2} ohm.....	23	25	26	30	40	45	49	70	100	145
Frequency in 10^3 cycles.....	3.50	6.46	15.3	21.5	50	75	100	125	150	
L in 10^{-6} henry....	581	578	574	572	570	568	568	570	572	
R in 10^{-2} ohm.....	245	495	1095	1345	2640	2940	3730	5280	7860	

Two-layer solenoids, one layer wound directly on the other, are not feasible for radio work, as the internal capacity is so high. Two-layer coils are sometimes used, however, the turns being arranged in a so-called "banked" winding. Multi-layer coils are, however, preferable, but they must be built in such a way as to keep the internal capacity low, as described on p. 164.

EQUATION (14) FOR ROUND WIRE
 b = dia. of round wire ~ per spiral
 d = gap between wires = dia. of wire used.

Inductances of a Flat Spiral, of Ribbon Conductors, Wound Flatwise, Turns Close Together.—

$$L = 4\pi R n^2 \left[\left(1 + \frac{3b^2 + d^2}{96R^2} \right) \log \frac{8R}{\sqrt{b^2 + d^2}} - C_1 + \frac{b^2}{16R^2} C_2 \right] \text{ cm., . (14)}$$



Coil #1 $D_1=14\frac{1}{2}"$ $D_2=17\frac{1}{2}"$ Turns = 29.5
 Coil #2 $D_1=14\frac{1}{2}"$ $D_2=16\frac{1}{2}"$ Turns = 34.5

FIG. 53.—Short solenoid made of edgewise-wound copper ribbon.

where R = mean radius of coil (see Fig. 54);

n = total number of turns;

b = width of strip = axial length of coil;

d = radial depth of coil = outside radius - inside radius.

C_1 and C_2 are constants depending on the shape of the spiral for their values. They are given in Table VII.

TABLE VII
 CONSTANTS C_1 AND C_2 FOR EQ. (14)

Ratio $\frac{b}{d}$	C_1	C_2
.00	.500	.125
.05	.549	.127
.10	.592	.133
.15	.631	.142
.20	.665	.155

Equation (14) gives the values obtained



FIG. 54

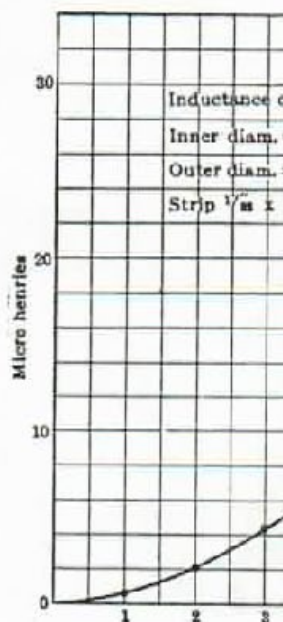


FIG. 55.—Inductance

per cent for the spacing. Fig. 55 shows how the inductance varies with the number of turns used.

Equation (14) gives incorrect values if the turns are not close together; the values obtained from the equation must be decreased as much as 5

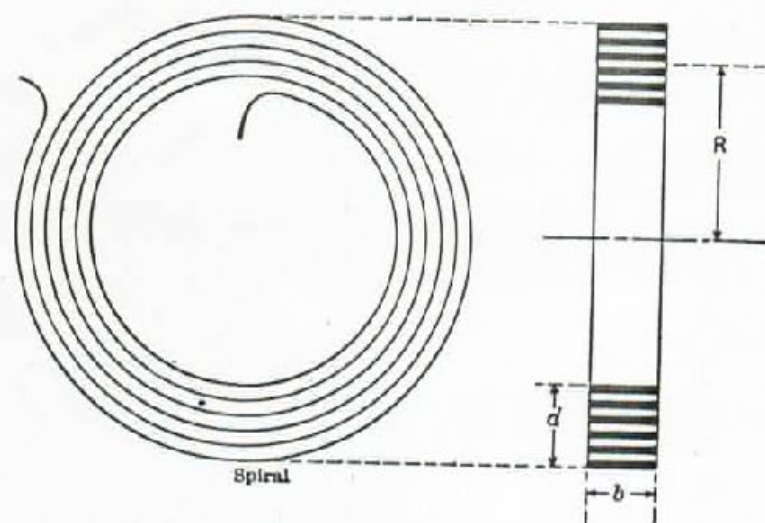


FIG. 54.—Spiral coil of ribbon wound flatwise.

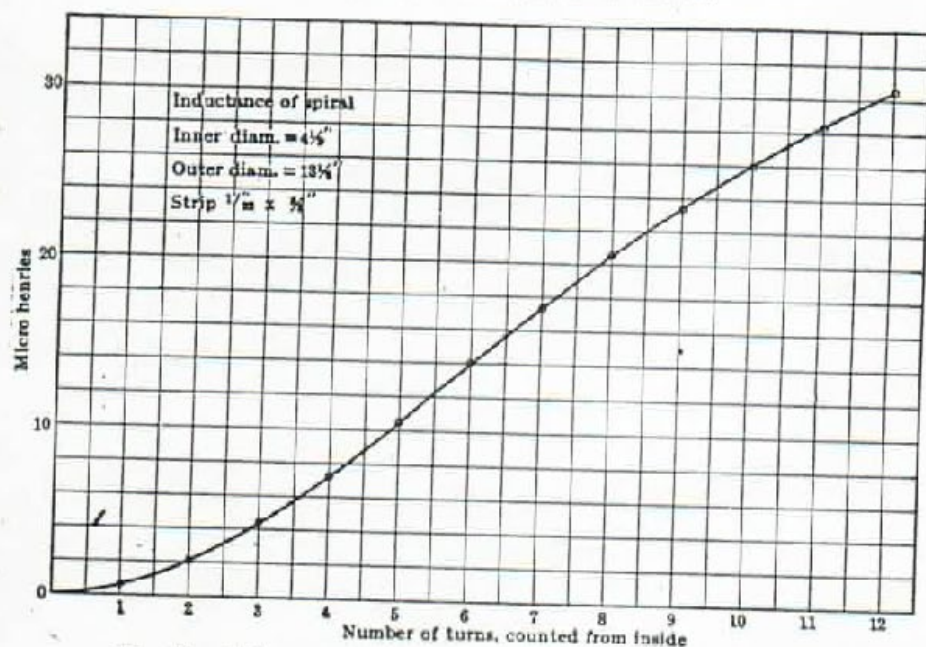


FIG. 55.—Inductance of a spiral similar to that shown in Fig. 54.

per cent for the spacing used in ordinary transmitting coils in spiral form. Fig. 55 shows how the value of L for a given spiral varies with the number of turns used.

It is interesting to note that the same length of wire will give about the same inductance whether wound into a flat spiral or a single-layer solenoid, provided that the mean radius of the spiral has the same value as the radius of the solenoid.

Toroidal Coil of Rectangular Cross-section. (Fig. 56.)—

$$L = 2n^2 l \log \frac{R_2}{R_1} \text{ cm.,} \quad (15)$$

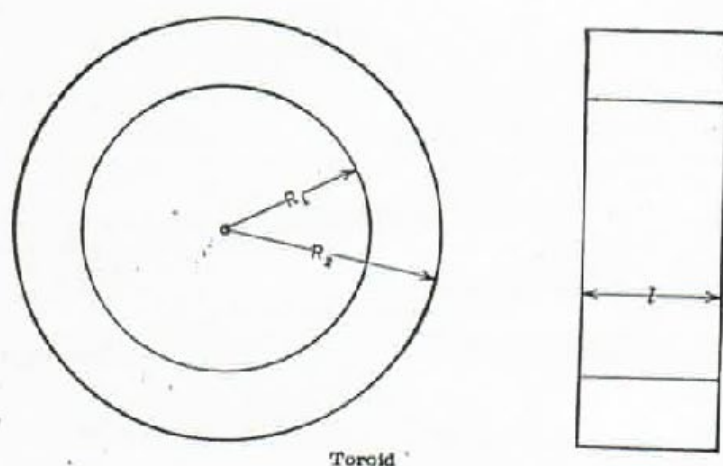


Fig. 56.—Toroidal coil of rectangular cross-section.

where n = total number of turns;
 l = axial length of coil;
 R_2 = outer radius;
 R_1 = inner radius.

Toroidal Coil of Circular Cross-section (Torus). (Fig. 57.)—

$$L = 4\pi n^2 (R - \sqrt{R^2 - r^2}) \text{ cm.,} \quad (16)$$

where n = total number of turns;
 R = mean radius of ring;
 r = radius of cross-section of winding.

The great advantage of a toroidal coil is that it has practically no external magnetic field and so gives but little mutual induction with other circuits. Also a toroidal coil will, for similar reasons, not be affected by mutual induction from other circuits or sources. Used as a tuning coil in a receiving set it will not pick up any strays or other disturbing fields unless they be of excessively short wave length.

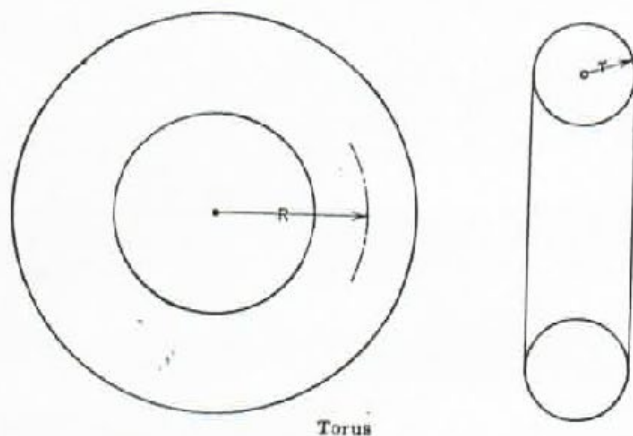
CO
 Single-Layer Squ
 $L = 8\pi n^2$



Fig. 58.—Single
 in which a = side of square
 n = number of turns
 D = pitch of winding
 b = axial length

Single-Layer Square Coil. (Fig. 58.)—

$$L = 8an^2 \left[\log \frac{a}{b} + .726 + .223 \frac{b}{a} \right] - 8an[A+B] \text{ cm.,} \quad (17)$$



Torus

FIG. 57.—Toroidal coil of circular cross-section.

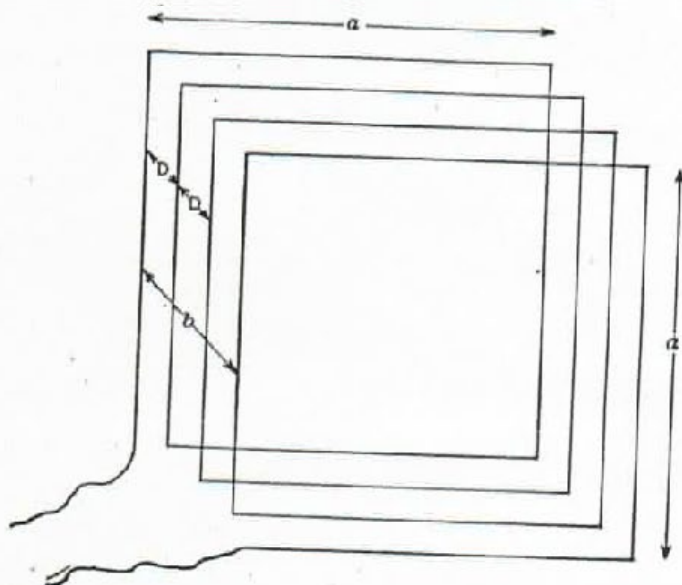


FIG. 58.—Single-layer square coil, such as is used for a coil antenna.

in which a = side of square, measured to center of wire;
 n = number of turns;
 D = pitch of winding, center to center;
 b = axial length of coil = $(n-1)D$.

A and B are constants depending upon number of turns, pitch, etc., and are given in Tables VIII and IX, d being the diameter of the wire used. Coils wound with rectangular conductor have slightly different constants than those given here.

TABLE VIII

$\frac{d}{D}$	A	$\frac{d}{D}$	A
1.00	+.557	.18	1.16
.90	.452	.16	1.28
.80	.334		
.70	.200	.14	1.41
.60	+.046	.12	1.56
.50	-.136	.10	1.75
.40	.356	.08	1.97
.35	.443	.06	2.26
.30	.647	.04	2.66
.25	.830	.02	3.36
.20	1.05		

TABLE IX

Number of Turns, n	B
1	.000
2	.114
3	.166
4	.197
6	.233
8	.253
10	.266
20	.297
40	.315
60	.322
100	.328

Flat Square Coil. (Fig. 59.)—

For this case Eq. (17) is applicable providing a is taken as $a_0 - (n-1)D$

where a_0 = side of square, outside wire;

n = number of turns;

D = distance between turns, center to center.

The value of b is obtained from the depth of the winding, i.e., it is equal to $(n-1)D$.

Multi-layer Coils of Rectangular Cross-section.¹ (Fig. 60.)—

$$L = \frac{(2\pi Rn)^2}{b+1.5t+R} F'F'' \text{ cm.,} \quad \dots \quad (18)$$

where R = mean radius of coil in cm.;

n = total number of turns in coil;

b = axial length of coil;

t = radial depth of winding.

¹ An excellent article on the design of multi-layer coils was published in Univ. of California Publications in Engineering, Vol. 147, by F. E. Pernot.

F' and F'' are cor

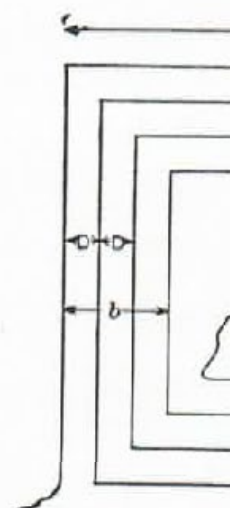


FIG. 59.—Flat sq

FIG. 60.—Multi-layer coil

For accurate resu
must be small compar

A multi-layer coil
ent, using a so-called
is shown in Fig. 61.



FIG. 61.—“

F' and F'' are correction factors

$$F' = \frac{10b + 13t + 2R}{10b + 10.7t + 1.4R}$$

$$F'' = 1.15 \log_{10} \left(100 + \frac{14R + 7t}{2b + 3t} \right)$$

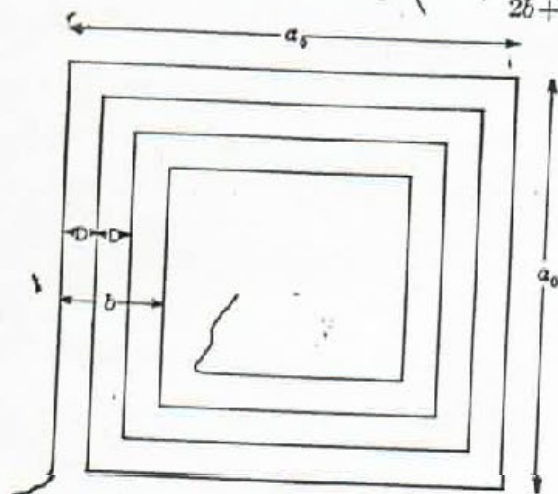


FIG. 59.

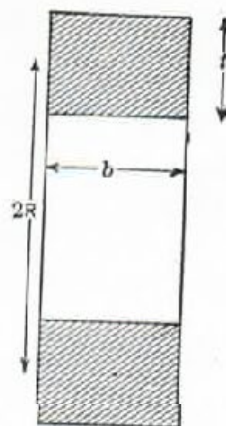


FIG. 60.

FIG. 59.—Flat square coil, used as coil antenna for short wave-lengths.

FIG. 60.—Multi-layer coil of rectangular cross-section; the cross-hatched area shows the cross-section of the winding.

For accurate results with this formula the distance between wires must be small compared to the diameter of the wire.

A multi-layer coil of very ingenious construction is being made at present, using a so-called honeycomb construction. A picture of such a coil is shown in Fig. 61. The coil is self-supporting, in this respect being

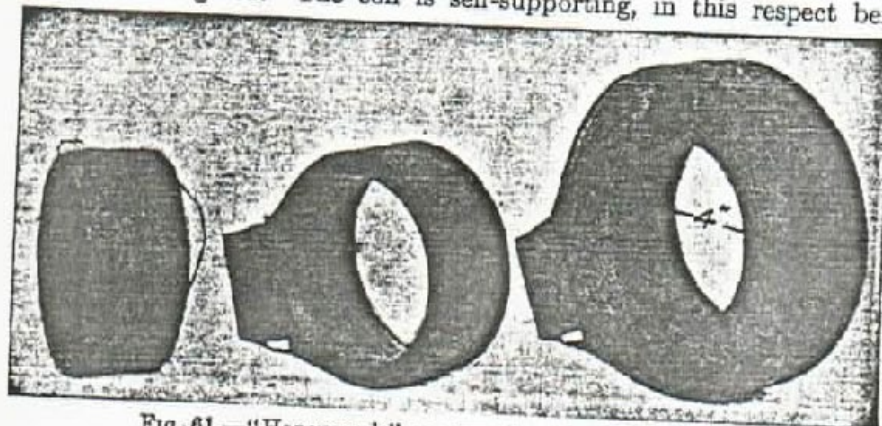


FIG. 61.—“Honeycomb” construction of multi-layer coil.

superior to the multi-layer coils described on p. 164 and although its internal capacity is greater than that of the type shown in Fig. 20, it is still sufficiently low to make it an excellent coil for radio circuits, especially those calling for many millihenries of inductance. The constants of one of these coils are shown in Table X.

TABLE X
CONSTANTS OF A HONEYCOMB COIL

Frequency in 10 ³ cycles	R in ohms	L in 10 ⁻⁴ henries	Reactance Divided by Resistance
0	9.39		
26.5	12.4	17.75	238
53	23.8	17.85	250
79.5	53.0	18.70	176
106	102.0	20.25	120

The dimensions of this coil were internal diameter=5 cm., external diameter=10 cm., cross-section of winding 2.5 cm. by 2.5 cm.

Most Efficient Form of Coil.—For a given length of wire it will be found that a certain form of coil gives a maximum inductance. Brooks and Turner have compiled a table to bring out the idea and some of their data is given in Table XI. In this table are shown constants of practically any form of coil into which the given length of wire could be wound.

TABLE XI
INDUCTANCE IN MILLIHENRIES OF 52.4 FEET OF MAGNET WIRE 0.1 IN. IN OUTSIDE DIAMETER (SMALL NO. 11 D.C.C.) WOUND INTO CYLINDRICAL COILS OF VARIOUS FORMS AS INDICATED.

(Dimensions are in Inches)

Description	Number of Layers	Total Turns	Mean Radius	Length of Coil	Thickness	Outside Radius	Inductance in Millihenries	Per Cent of Maximum
Spaced solenoid.....	1	80	1.25	16.0	0.1	1.3	.058	17
Solenoid.....	1	80	1.25	8.0	0.1	1.3	.108	33
Double-layer solenoid.....	2	80	1.25	4.0	0.2	1.35	.186	56
Thick tube.....	4	80	1.25	2.0	0.4	1.45	.279	84
Compact.....	8	80	1.25	1.0	0.8	1.65	.331	100
Thick disk.....	16	80	1.25	0.5	1.6	2.05	.289	87
Thick disk.....	10	50	2.0	0.5	1.0	2.5	.301	91
Square section ring.....	5	25	4.0	0.5	0.5	4.25	.244	74
Flat ring.....	4	8	12.5	0.2	0.4	12.7	.119	36
Thin disk.....	2	2	50.0	0.1	0.2	50.1	.042	13
Single turn.....	1	1	100.0	0.1	0.1	100.0	.025	8

As the same length are all the same. It is only 33 per cent as good.

However, it must be much greater (at radio frequencies) because the increase in resistance is well brought out in Table XII. A 12 wire, single layer, inductance from zero to 150 kilohms the same wire, in the same times.

With this idea in mind, resistance (which is really better for the single layer) serves well to illustrate the difference at radio frequencies flowing through it.

Variable Inductance.—Inductance is obviously variable inductance. If used, one a long solenoid coils connected in series, this type being generally.

The solenoid with contact frequently lies short-circuited turn, de its proper value for the in the short-circuited turn. Also there is not much method; for long solenoid of the length only and contact set to connect in or occurring in the long u called a "dead end") is former; the charging current and so increases the efficiency used. Long solenoids in into a number of complete series as required.

The variometer type being continuously variable is shown in Fig. 62, from seen. If the ends of the out to separate terminals

As the same length of wire is used in all these coils their c.c. resistances are all the same. It would seem then that the single-layer solenoid is only 33 per cent as good as a banked coil having eight layers.

However, it must be remembered that the a.c. resistance is generally much greater (at radio frequencies) than the c.c. resistance, and that this increase in resistance is greater as the coil is more compact. This effect is well brought out in Tables II and III, which indicate that a coil of No. 12 wire, single layer, increases its resistance 12 times in the frequency range from zero to 150 kilocycles, whereas a square section coil (10 layers) of the same wire, in the same frequency range, increases its resistance 170 times.

With this idea in mind we must conclude that the ratio of reactance to resistance (which is really the figure of merit for a radio coil) would probably be better for the single-layer solenoid than for the compact coil. This serves well to illustrate the fact that simple apparatus may behave quite differently at radio frequency than it does when continuous current is flowing through it.

Variable Inductances.—It is many times desirable to have a continuously variable inductance for tuning a circuit; two such types have been used, one a long solenoid with a sliding contact and the other a pair of coils connected in series, one rotatable inside the other, an inductance of this type being generally styled a variometer.

The solenoid with sliding contact is not good, because the sliding contact frequently lies on two turns at the same time, thus producing a short-circuited turn, decreasing very appreciably the self-induction from its proper value for the position of the contact, and, due to the current in the short-circuited turn, increasing the effective resistance of the coil. Also there is not much useful variation of inductance obtainable by this method; for long solenoids the value of L increases with the first power of the length only and the coil cannot be used effectively with the contact set to connect in only a small portion of the coil because of the losses occurring in the long unused portion. This part of the coil (generally called a "dead end") is excited like the secondary of a step-up auto transformer; the charging current circulating in the dead end produces losses and so increases the effective resistance of that part of the coil which is used. Long solenoids intended to be used in steps should be divided up into a number of completely insulated sections, these being connected in series as required.

The variometer type of inductance is very convenient and useful, it being continuously variable; the calibration curve of such an inductance is shown in Fig. 62, from which the probable range in inductance can be seen. If the ends of the stationary coil and rotating coil are brought out to separate terminals, the combination forms a very convenient

scheme of magnetically coupling two independent circuits, a so-called "coupler."

A rather convenient scheme (even though inefficient) for making a continuously variable inductance out of a short solenoid, is to fix a copper disk on a shaft inside the solenoid, so that the axis of the disk may be made parallel or not to that of the coil. The eddy currents in the disk, with parallel axes, will very materially reduce the inductance of the solenoid.

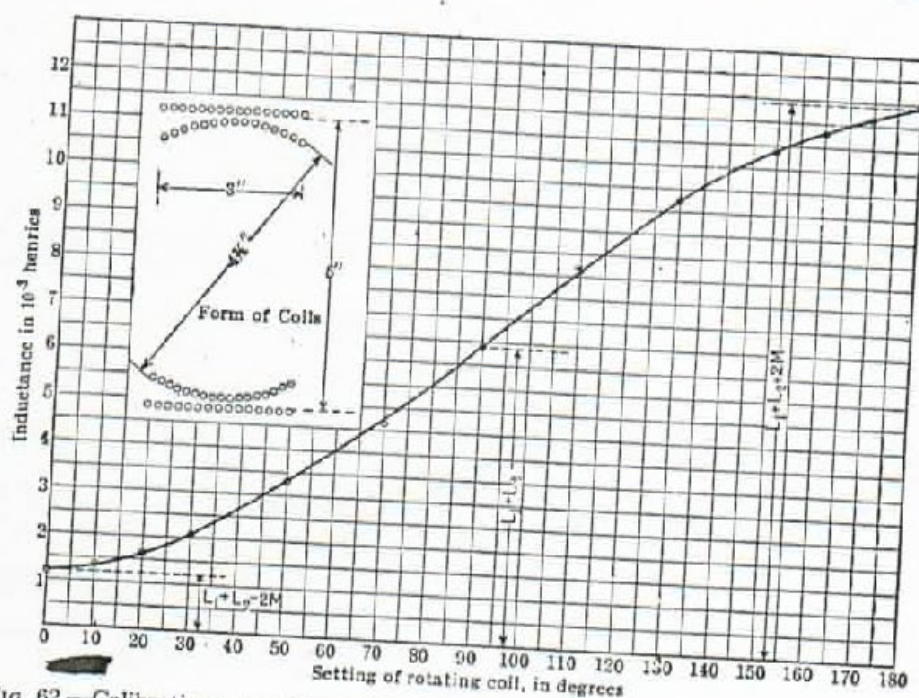


FIG. 62.—Calibration curve of a variable inductance commonly known as a variometer.

The effect of such a solid disk placed inside a short solenoid is given in Table XII. The coil was a single-layer solenoid 12 cm. in diameter and of 2 cm. axial length; the various disks were 11 cm. diameter and were placed inside the coil, centrally, with the plane of the disk perpendicular to the axis of the coil. It is evident that a copper disk very materially affects the inductance without prohibitive increase in resistance. By having the plane of the disk rotatable this scheme of varying the inductance of a coil may be useful, e.g., in heterodyne reception, where but slight changes in inductance are desired (to change signal note), and an increase in resistance of the coil is not of serious consequence.

The best adjustable inductance is a multi-layer coil, with each layer (or every other one after the first four perhaps) separate from the others,

equipped with the proper as desired.

Frequency in Kilocycles	EFFECT	
	Coil alone	
	L 10 ⁻³ henry	R ohms
1	1.069	3.03
5.35	1.052	3.20
50	1.058	3.35
149	1.092	5.08

MU

Mutual Induction.—The may be expressed in terms If two coils, so situated with netic field of each is linked series in such a way that t with the magnetic field of t trical connection is reverse circuit which changes that a corresponding effect on tl considered as fixed by the other as given in Eq. (6), C

The M of the two coils may be changed, however, tive position of the two coi third circuit is brought into of the two coils. Thus t coils, placed with their en may have a value of M ab L_1 , but if a copper sheet is the two coils the value of Eq. (6) of Chapter I may l to zero at high frequencies.

The values of M for a rangements are given below

equipped with the proper switch to connect in the circuit as many layers as desired.

TABLE XII
EFFECT OF METAL DISK INSIDE SOLENOID

Frequency in Kilocycles	Coil alone		$\frac{1}{4}$ -in. Copper Disk		$\frac{1}{4}$ -in. Brass		$\frac{1}{4}$ -in. Brass		$\frac{1}{4}$ -in. Tinned Iron	
	L 10 ⁻³ henry	R ohms	L	R	L	R	L	R	L	R
1	1.069	3.63	.807	3.49	.895	3.77	1.025	3.53	1.055	3.25
5.35	1.052	3.20	.762	4.13	.785	5.13	.870	6.40	.970	6.40
50	1.058	3.35	.751	5.79	.756	8.13	.783	12.6	.865	18.6
149	1.092	5.08	.760	9.17	.765	13.5	.792	17.8	.861	44.3

MUTUAL INDUCTION

Mutual Induction.—The coefficient of mutual induction of two coils may be expressed in terms of energy in the same way as is self-induction. If two coils, so situated with respect to one another that part of the magnetic field of each is linked with the other, are connected electrically in series in such a way that their m.m.f.s add, the total energy associated with the magnetic field of the circuit is $\frac{1}{2}I^2(L_1+L_2+2M)$ and if the electrical connection is reversed, it is $\frac{1}{2}I^2(L_1+L_2-2M)$. Any change in the circuit which changes that portion of the magnetic energy due to M has a corresponding effect on the value of M . The value of M may also be considered as fixed by the voltage induced in one coil by current in the other as given in Eq. (6), Chapter I.

The M of the two coils is determined by their relative position; it may be changed, however, even if the relative position of the two coils stays fixed, if a third circuit is brought into the mutual field of the two coils. Thus two equal coaxial coils, placed with their ends close together may have a value of M about .7 as large as L_1 , but if a copper sheet is inserted between the two coils the value of M as defined by Eq. (6) of Chapter I may be brought nearly to zero at high frequencies.

The values of M for a few ordinary arrangements are given below, the formulæ being approximations as were

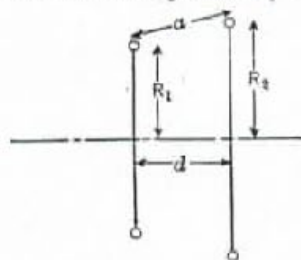


FIG. 63.—Cross-section through two single turns placed coaxially.